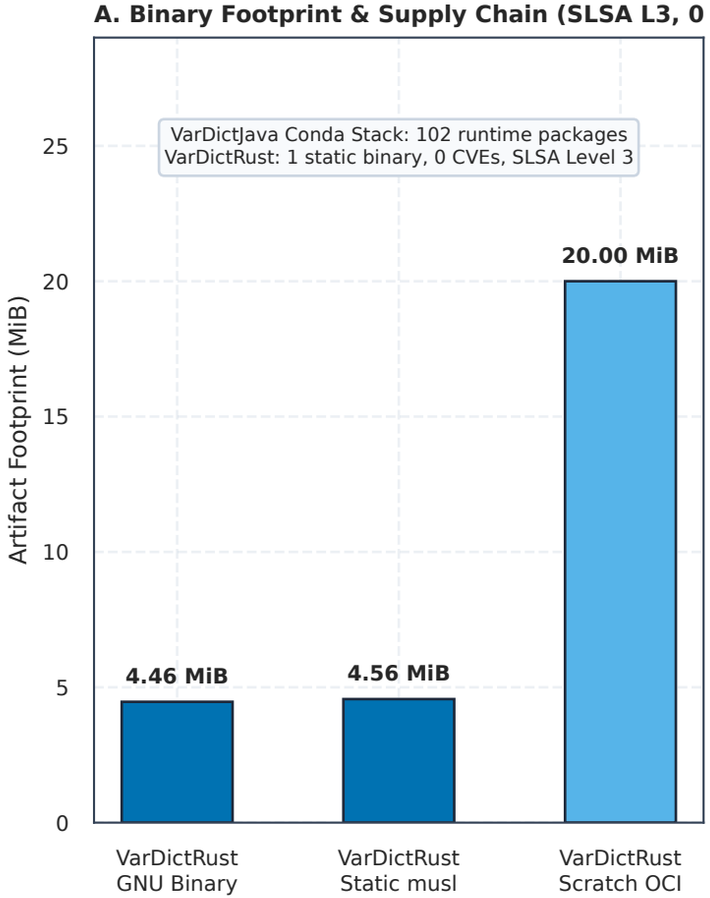
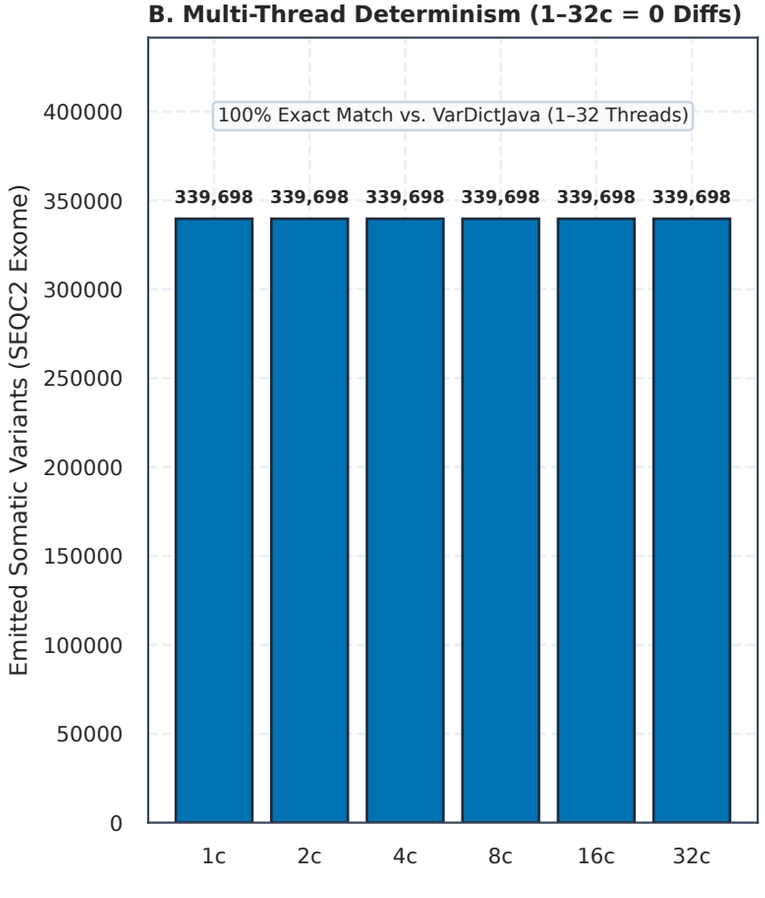
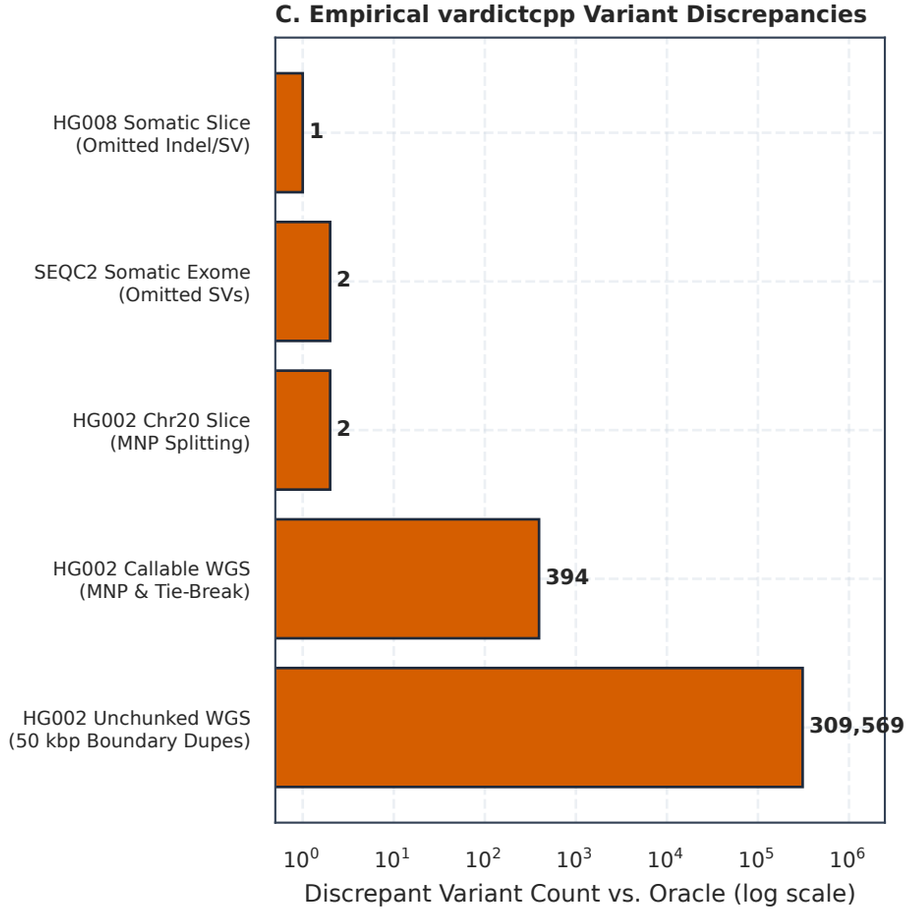

A. Binary Footprint & Supply Chain (SLSA L3, 0 CVEs)
VarDictJava Conda Stack: 102 runtime packages
VarDictRust: 1 static binary, 0 CVEs, SLSA Level 3
Artifact Footprint (MiB)
0
5
10
15
20
25
4.46 MiB
4.56 MiB
20.00 MiB
VarDictRust GNU Binary
VarDictRust Static musl
VarDictRust Scratch OCI


B. Multi-Thread Determinism (1–32c = 0 Diffs)
100% Exact Match vs. VarDictJava (1–32 Threads)
Emitted Somatic Variants (SEQC2 Exome)
0
50000
100000
150000
200000
250000
300000
350000
400000
339,698
339,698
339,698
339,698
339,698
339,698
1c
2c
4c
8c
16c
32c


C. Empirical vardictcpp Variant Discrepancies
HG008 Somatic Slice (Omitted Indel/SV)
1
SEQC2 Somatic Exome (Omitted SVs)
2
HG002 Chr20 Slice (MNP Splitting)
2
HG002 Callable WGS (MNP & Tie-Break)
394
HG002 Unchunked WGS (50 kbp Boundary Dupes)
309,569
10^0
10^1
10^2
10^3
10^4
10^5
10^6
Discrepant Variant Count vs. Oracle (log scale)